INCH-POUND
MIL-D-70860 (AR)
14 February 1992

SUPERSEEDING
(See 6.9)

MILITARY SPECIFICATION

DETECTOR BELT ASSEMBLY, GENERAL SPECIFICATION FOR

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture, inspection, packaging and acceptance of the Detector Belt Assembly.

Beneficial comments (recommendations, addition, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self- addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1265

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	- Fire Control Materiel Manufacture and Inspection of, General Specification for
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-W-63150	- Weapons and Support Materiel, Standard Quality Assurance Provisions for

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	- Reliability Tests, Exponential Distribution
MIL-STD-810	- Environmental Test Methods
MIL-STD-1686	- Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding electrically initiated explosive devices)
MIL-STD-1695	- Environments, Working, Minimum Standards for

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MIL-STD-2000 - Standard Requirements for Soldered Electrical and Electronic Assemblies

MIL-STD-45662 - Calibration Systems Requirements

HANDBOOK

MILITARY

DoD-HBK-263 - Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding electrically initiated explosive devices) METRIC

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER.

PRODUCT DRAWINGS (CAGE CODE 19200)

11749230	-	Detector Belt Assy, Segment No. 1 - CVLD
11749231	-	Electronics Module Assy, CVLD
11749238	-	Detector Belt Assy, Segment No. 2 - CVLD
11749241	-	Connection Diagram, Detector Belt Segment No. 1
11749296	-	Detector Belt Assy, Segment No. 4 - CVLD
11749300	-	Detector Belt Assy, Segment No. 3 - CVLD
11835890	-	Detector Belt Assy, Segment No. 5 - CVLD
11836200	-	Detector Belt Assy, Segment No. 6 - CVLD

INSPECTION EQUIPMENT (CAGE CODE 19200)

11749279	-	Controller Gun (Ref.) (6.8)
11749549	-	Console, Simulator System, Laser M60 Tanks and M551 Vehicle (Ref.) (6.8)

PACKAGING DATA SHEET (CAGE CODE 19200)

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SPI-P11749230	-	Detector Belt Assy, Segment No. 1 - CVLD
SPI-P11749238	-	Detector Belt Assy, Segment No. 2 - CVLD
SPI-P11749296	-	Detector Belt Assy, Segment No. 4 - CVLD
SPI-P11749300	-	Detector Belt Assy, Segment No. 3 - CVLD
SPI-P11835890	-	Detector Belt Assy, Segment No. 5 - CVLD
SPI-P11836200	-	Detector Belt Assy, Segment No. 6 - CVLD

(Copies of other Government documents, drawings and publications required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item description. The MILES Detector Belt Assembly segments are designed for use in predetermined combinations which comprise the Combat Vehicle Laser Detection (CVLD) System. The CVLD is used to detect "HIT/NEAR MISS" conditions when installed on vehicles in the MILES system as defined in 6.1. Each belt is constructed of heavy nylon webbing and contains Si-PN Laser Detector Modules and Electronic Threshold/Detector Preamplifier Module which interconnects to the companion decoder through adapter cables. The belt segments differ primarily in length and the location and quantity of the laser detectors. All wiring between components is sewn internally in the nylon webbing. The belt segments are powered by the companion decoder and are quickly attached/detached by use of velcro fasteners.

3.2 First article. When specified (see 6.4), a sample shall be subjected to first article inspection in accordance with 4.4.

3.3 Environmental stress screen (ESS). All assemblies shall be subjected to an environmental stress screen prior to conformal coating consisting of temperature cycling and soaking between the temperatures -35°C and +62.8°C. After exposure to ESS in accordance with 4.5, the characteristics of 3.4 shall not be degraded.

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3.4 Charactristics.

3.4.1 Operating conditions. The operating conditions shall be in accordance with Table I, measured under quiescent conditions. For terminal and component designations called out in this specification, see drawing 11749241, Connection Diagram, Detector Belt, Segment No. 1-CVLD.

TABLE I. Operating Conditions.

Parameter	Terminal	Minimum	Nominal	Maximum	Unit
Ambient Temperature		-31.7	25	62.8	°C
Input (+) Power	P1, pin A	7.3	11	13	Vdc
Return (-) Power	P1, pin B				
Input Current	P1, pin A			3.0 5.0	mA mA 1 /

Note:
1 / Detectors receiving laser pulses.

3.4.2 Performance. Unless otherwise specified, the PWBA shall meet the requirements of this specification under the following conditions:

a.	Temperature	25°±10°C (77°±18°F)
b.	Relative humidity	50±30 %
c.	Barometric pressure	725+50, -75mm Hg (28.5+2.0, -3.0 inches Hg)
d.	Input power and loads	See Figure 1

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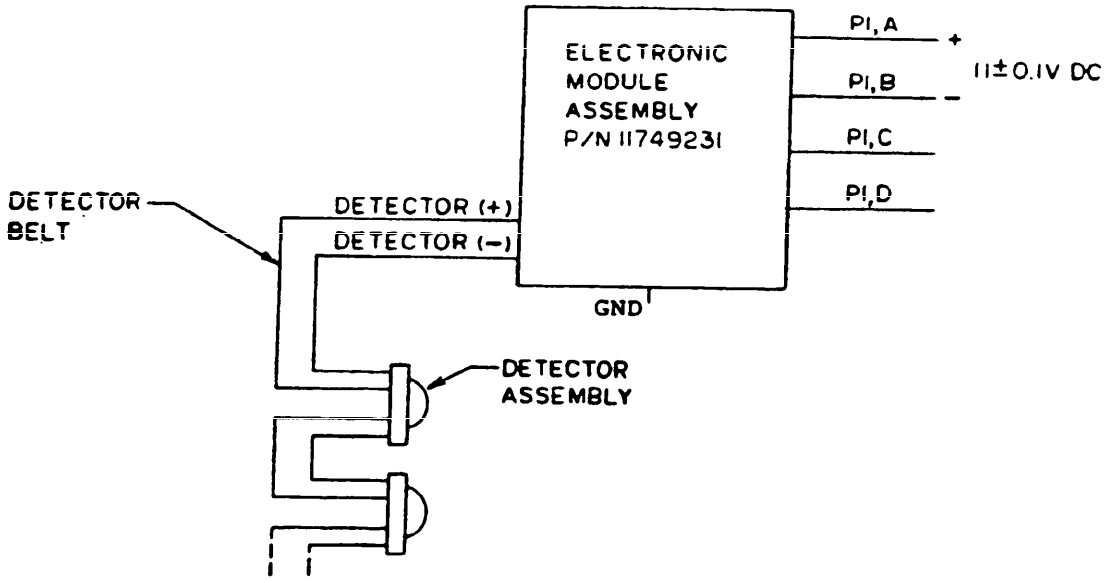


FIGURE 1. Detector Belt Power Source and Loads.

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3.4.2.1 Current drain. With power applied, the quiescent current drain shall be less than 3.0 mA.

3.4.2.2 Detection sensitivity. The detector belt assemblies, when powered, shall detect GaAs laser radiation in the range of 880 to 920 nm impinging upon the belt assembly. The detector belt segments shall operate in ambient illumination ranging from darkness to full sunlight and shall detect GaAs laser pulses with total collected energy given in Table II. The detected laser light shall generate an output at pins C (+) and pin D (-) of terminal P1. The detected light pulses shall be represented by active low bits (see Figure 2).

TABLE II. Laser Light Energy.

on Assembly Identification			Energy Threshold- one Detector 1/4/ (Microergs)	
Part No.	Description	No. of Detectors	Nominal	Max Allowable on 1/2 or less of Total Detectors
11749230	Segment NO. 1	6	28	34
11749238	Segment No. 2	4	22	28
11749300	Segment No. 3	8	34	43
11749296	Segment No. 4	6	28	34
11835890	Segment No. 5	14		
		(V1 thru V6)	28	34 2/
		(V7 thru V14)	34	43 3/
11836200	Segment No. 6	6	28	34

NOTES:
1/ Up to 50 percent of the total number of detectors may have the higher energy requirement. In the case of Segment No. 3, up to three detectors may require the higher energy.
2/ Three detectors may require the higher threshold.
3/ Four detectors may require the higher threshold.
4/ Individual detection modules shall be additive with regard to received laser energy so that weaker fields may be detected if they impinge upon more than one detector.

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FIGURE 2. Output Signal.

3.4.2.3 Grounding. The maximum circuit resistance between adjacent detector housings and the electronic module (GND) shall be 20 mohms. The maximum circuit resistance between adjacent detector housings shall be 20 mohms. The maximum circuit resistance between the electronic module (GND) and its associated output connector (P1, C) shall be 80 mohms.

3.4.3 Environmental requirements. Unless otherwise specified the assembly shall be subjected to the following environmental conditions and meet the stated requirements.

3.4.3.1 High temperature. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2. when exposed to and after the following.

3.4.3.1.1 Operating temperature. The assembly shall operate up to a temperature of +62.8°C(+145°F) after exposure at +62.8°C for 4 hours.

3.4.3.2 Low temperature. The assembly shall not be damaged

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and shall meet the performance requirements of 3.4.2 when exposed to and after the following.

3.4.3.2.1 Operating temperature. The assembly shall operate down to a temperature of -34.4°C(-30°F).

3.4.3.2.2 Storage temperature. The assembly shall operate after exposure to a temperature of -31.7°C(-25°F) for a period of 24 hours.

3.4.3.3 Shock. The assembly shall meet the performance requirements of 3.4.2 after shocks with a nominal level of 40 g's terminal peak and nominal duration of 11 milliseconds as shown in Figure 3. Apply three shocks (18 total) in each direction of the three mutually perpendicular axes.

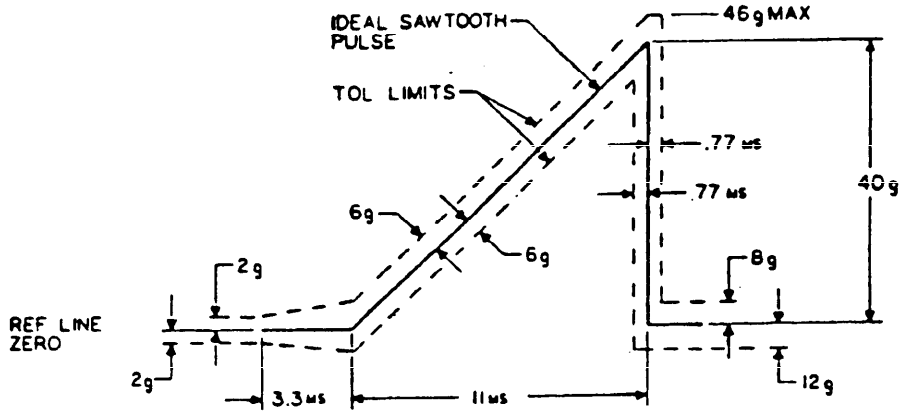


FIGURE 3. Shock Pulse.

3.4.3.4 Vibration. The assembly shall meet the performance requirements of 3.4.2 after sinusoidal vibrations of frequency and amplitude given below and in Figure 4 on three mutually perpendicular axes. Total duration shall be 60 minutes per axis.

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Frequency	Amplitude	Sweep rate
7-30 Hz	±1.5 g	7-500-7 Hz
30-50 Hz	0.033" DA	in 15 minutes
50-500 Hz	±4.2 g	

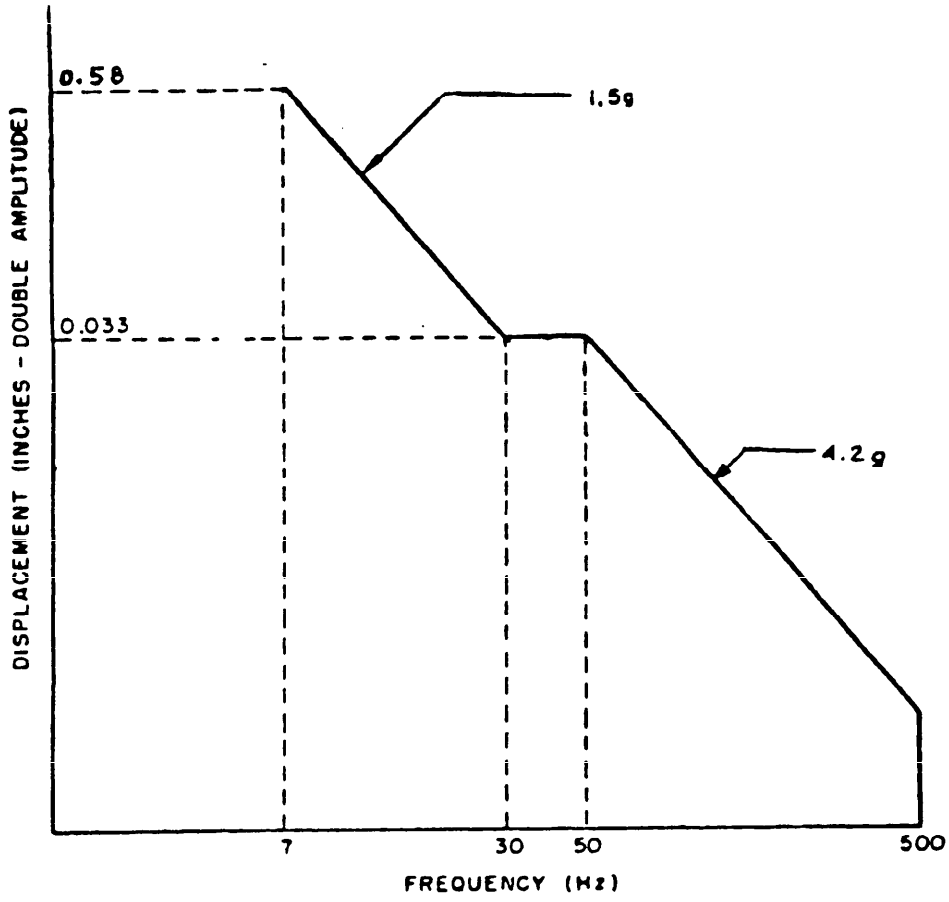


FIGURE 4. Vibration Test.

3.4.3.5 Low pressure. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2 when exposed to a barometric pressure of 565 TORR.

3.4.3.6 Humidity. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2 after exposure to humidity in a natural cyclic environment. The natural cyclic

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humidity environment shall consist of ten 24 hour cycles in accordance with the temperature-humidity requirements of Figure 5.

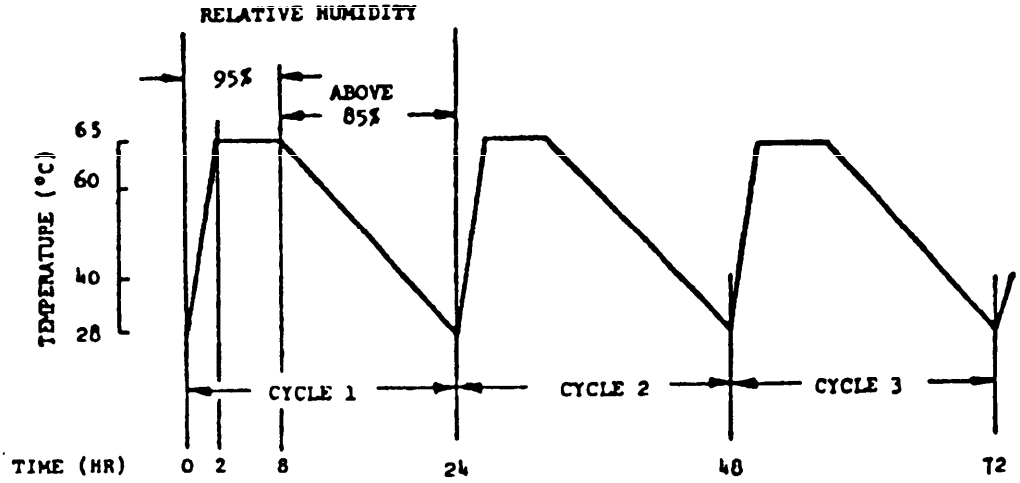


FIGURE 5. Humidity Cycle.

3.4.3.7 Rain. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2 after exposure to blowing rain. The blowing rain shall last 30 minutes according to the following requirements:

a. Simulated rain at a rate of 2 ± 0.5 inches per hour for 10 minutes with the water temperature, for the full 30 minutes, between 52°F and 95°F.
b. Simulated rain at a rate of 5 ± 0.5 inches per hour from 10 to 15 minutes from the initiation of the rain.
c. Simulated rain at a rate of 2 ± 0.5 inches per hour form 15 to 30 minutes from the initiation of the rain.
d. Blowing horizontal wind at a velocity of 40 miles per hour starting 5 minutes from the initiation of the rain and lasting for 15 minutes.

3.4.3.8 Dust. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2 after exposure to blowing dust. The blowing dust environment shall consist of 3 phases over a 28 hour period according to the following requirements:

a. Expose the assembly for 6 hours at a temperature of 23°C (73°F), with a relative humidity of less than 22%

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(for all phases), to a dust concentration of 0.3 ± 0.2 grams/cubic feet and an air velocity of 1750 ± 250 feet/minute.

b. Expose the assembly for 16 hours, at a temperature of 62.8°C (145°F), to no dust concentration and an air velocity of 300 ± 200 feet/minute.

c. Expose the assembly for 6 hours at a temperature of 62.8°C (145°F) to a dust concentration of 0.3 ± 0.2 grams/cubic feet and an air velocity of 1750 ± 250 feet/minute.

3.4.3.9 Salt fog. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2 after exposure to salt fog. The salt fog shall last 48 hours according to the following requirements:

a. Salt solution of 5%.
b. Salt solution pH between 6.5 and 7.2.
c. Air temperature of 35°C.

3.4.3.10 Electromagnetic Interference (EMI). The assembly shall not be damaged and shall meet the performance requirements of 3.4.2 when exposed to the following tests of MIL-STD-461, except as modified below.

Test	Modification
RE02	Broadband limits shall be increased by 8 dB V/MHz at all frequencies. Narrowband limits shall be increased by 14 dB V at all frequencies.
RS01	-
RS03	10 KHz to 8.9 MHz - 1 V/m 9.0 MHz to 400 MHz - 5 V/m

3.5 Design and construction. The PWBA shall be fabricated and assembled in accordance with drawings, parts lists, and other documents specified on drawings 11749230, 11749238, 11749296, 11749300, 11835890 and 11836200. The PWBA shall be manufactured and tested in a working environment in accordance with Table I, work area 5 of MIL-STD-1695.

3.5.1 Standards of manufacture. Handling of Electrostatic Sensitive Parts. This equipment contains semiconductor devices such as CMOS, mosfets, and microelectronic devices which may be

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degraded or damaged by an inadvertent discharge of static electricity. To prevent damage to these components, the supplier shall provide for adequate handling procedures and equipment in accordance with MIL-STD-1686 and DoD-HBK-263.

3.5.2 Soldering. Soldering shall be in accordance with the general requirements of section 4 of MIL-STD-2000.

3.5.3 Workmanship. General workmanship shall be in accordance with requirement 9 of MIL-STD-454.

3.6 Reliability. The detector belt assembly shall meet the following quantitative reliability requirements, in terms of mean-time-between failures (MTBF).

3.6.1 Specified value. The belt assembly shall have a MTBF value of 8000 hours.

3.6.2 Minimum acceptable value. The belt assembly shall have a minimum acceptable MTBF value of 4000 hours.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept

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or request for proposal, form ... Date of invitation for bids

defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Environmental stress screen (see 4.5).

c. Quality conformance inspection (see 4.6).

d. Packaging inspection (see 4.6.4).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the working environment specified in 3.5 and at standard ambient conditions defined in 3.4.2 using the methods of 4.8 and 4.9.

4.4 First article inspection.

4.4.1 Sample. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.4.2. The first article sample shall consist of three printed wiring board assemblies.

4.4.2 Inspections to be performed. First article inspections are listed in Tables III and IV. Each first article sample unit shall be inspected in accordance with the Table III before being subjected to the environmental tests in Table IV in the sequence listed. As determined by the Government, the first article assemblies may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.4.3 Rejection. If any assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly to comply with any of the requirements.

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TABLE III. Examinations and Performance Tests. 1/ 2/

Class	Characteristics	Requirements Paragraph	Test/ inspection Method	First Article Sample Size	Quality 3/ Conformance Inspection Sample Size/ Accept Criteria
101	Current Drain	3.4.2.1	4.8.1.1	3	32/Acc-0,Rej-1
102	Detector Sensitivity	3.4.2.2	4.8.1.2	3	32/Acc-0,Rej-1
103	Grounding	3.4.2.3	4.8.1.3	3	32/Acc-0,Rej-1
201	Workmanship	3.5.3	Visual	3	125/Acc-0,Rej-1
202	Soldering 4/	3.5.2	Visual	3	125/Acc-0,Rej-1
203	Dim Characteristics	3.5	SME 5/	3	125/Acc-0,Rej-1
204	Comp Orientation	3.5	Visual	3	125/Acc-0,Rej-1
205	Comp and Board Identification	3.5	Visual	3	125/Acc-0,Rej-1

NOTES:
1/ ESS testing per 4.5 is required for all deliverable units.
2/ 200 class (minor characteristics) examinations required prior to 100 class (major characteristics) performance tests.
3/ For lot sizes less than 32 or 125 as listed, 100% inspection is required (see 4.6.5).
4/ Check before conformal coating is applied.
5/ Standard Measuring Equipment.

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TABLE IV. Environmental Tests.

Class	Characteristic	Requirements Paragraph	Test Paragraph	First Article Sample Size	Quality Conformance Inspection Sample Size/ Acceptance Criteria
104	High Temp	3.4.3.1	4.9.1	3	1/Acc-0, Rej-1
105	Low Temp	3.4.3.2	4.9.2	3	1/Acc-0, Rej-1
106	Shock	3.4.3.3	4.9.3	3	1/
107	Vibration	3.4.3.4	4.9.4	3	1/
108	Low Pressure	3.4.3.5	4.9.5	3	1/
109	Humidity	3.4.3.6	4.9.6	3	1/
110	Rain	3.4.3.7	4.9.7	3	1/
111	Dust	3.4.3.8	4.9.8	3	1/
112	Salt Fog	3.4.3.9	4.9.9	3	1/
113	Electromagnetic Interference	3.4.3.10	4.9.10	3	1/

NOTE:
1/ Not required for quality conformance inspection.

4.5 Environmental stress screen. An environmental stress screening program shall be established. All units shall be subjected thermal stress screen in accordance with table V and the procedure below.

4.5.1. Conformal coating. Thermal stress screen shall be performed prior to conformal coating unless another method is approved by the Government.

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TABLE V. ESS Temperature Cycling Requirements.

Low Temp	High Temp	Rate Change	Power	Monitor	Cycles/Group A Test
-34.4°C	+62.8°C	15°C/minute	Off	No	20/Failure Free or Additional Cycle

4.5.2 Chamber temperature. Chamber temperature cycling shall be from -34.4°C (-30°F) to +62.8°C(+145°F) and the duration of one cycle to be 8 minutes up and 8 minutes down plus 4 minutes dwell at high temperature and 4 minutes at low temperature for a total cycle duration of 24 minutes. The units shall begin and end the cycles at ambient temperature (See Figure 6).

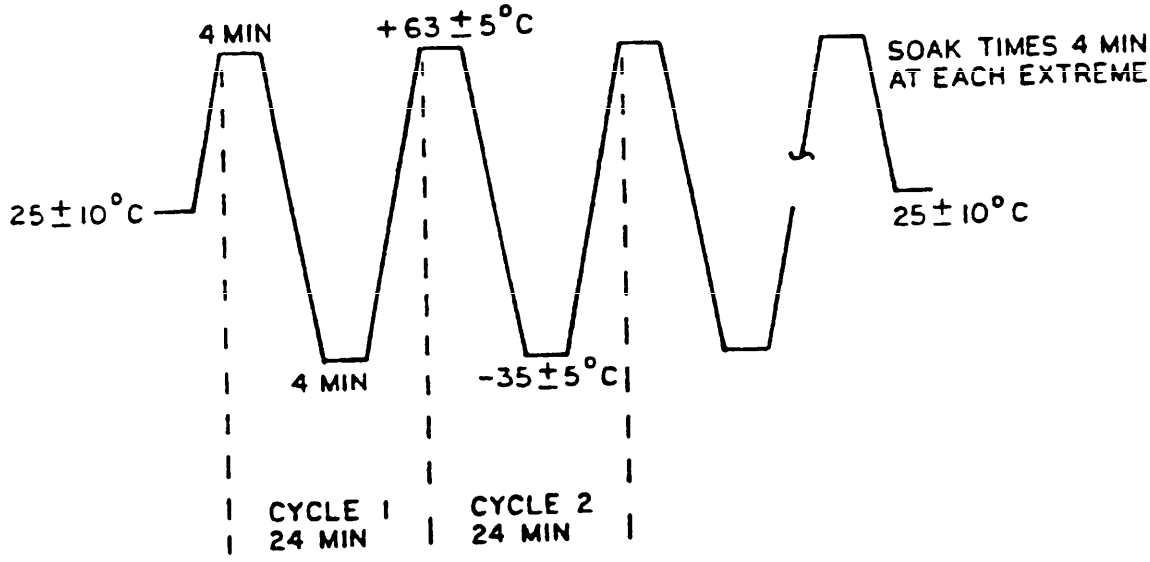


FIGURE 6. Environmental Stress Screen Temperatures Profile.

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4.5.3 Cycle frequency. A total of twenty thermal cycles are to be imposed on each unit.

4.5.4 Temperature cycles. Temperature cycling shall be conducted nonoperating and without power (static).

4.5.5 Test requirements. After the thermal stress screen, the units shall meet the Group A tests (4.6.2). If a unit is rejected, the unit shall be repaired for the failure (if possible) and be put through one additional cycle of ESS. The unit shall then undergo and pass Group A tests.

4.6 Quality conformance inspection. Quality conformance inspection requirements are defined in 4.6.2 and 4.6.3.

4.6.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn of which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior of subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.6.2 Group A tests. Group A tests consist of examinations and performance tests listed in Table III in the quantities specified.

4.6.3 Group B tests. Group B tests shall consist of the tests indicated in Table IV and shall be conducted on one sample unit per inspection lot which has been subjected to, and passed, Group A tests (see 4.6.2).

4.6.4 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall verify that the units are packaged in accordance with the requirements of 5.1. During the period of first article inspection, one additional unit, packaged in accordance with 5.1, shall be

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examined to the requirements for quality conformance inspection listed below. Failure of packaging shall be cause for rejection of packaging. Two units per lot shall be examined for the following:

a. Verify that the outside bag is marked in accordance with MIL-STD-129 and is properly closed.

b. Verify that sleeve is around box and that foam properly surrounds inner bag.

c. Verify that the inner bag is properly sealed.

d. Verify that the unit is not damaged after the removal of all packaging materials.

4.6.5 Failure of sample. Should any one sample fail to meet any of the specified examinations and test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract (see 6.7).

4.7 Inspection equipment.

4.7.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607 and MIL-W-63150.

4.7.2 Contractor-furnished inspection equipment.

4.7.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAP/QARs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs (see 6.3).

4.7.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the Methods of Inspection specified in 4.8 and 4.9 of this specification and with the component inspection procedures specified in Examination and Test Facilities requirements of MIL-F-13926, whenever Government designs are not specified (see 4.7.2.1). Since tolerance of test equipment is normally considered to be within

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ten percent of product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. When specified in contract or purchase order, the contractor shall submit inspection equipment designs for Government review and evaluation (see 6.3).

4.7.2.3 Test equipment. In conjunction with 4.7.2.1 and 4.7.2.2, the test equipment listed in Table VI (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.8.

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TABLE VI. Test Equipment List. 1/

Power Supplies (DC)	Standard - capable of providing levels and limits stated in 4.8.1.
DC Voltmeter/Multimeter	Standard - capable of measuring voltages stated in 4.8.1.
Ammeter/Multimeter	Standard - capable of measuring currents stated in 4.8.1.
Oscilloscope	Standard - capable of measuring levels stated in 4.8.1.
Controller Gun	P/N 11749279 - listed in 4.8.1.2
Ohmmeter/Multimeter	Standard- capable of measuring levels stated in 4.8.1.
Laser Power Meter and Pulse Integrator	EG&G 460-1A and 460-3 or equivalent - capable of measuring levels stated in 4.8.1.
Laser Detector Head	EG&G 460-2 or equivalent - capable of measuring levels stated in 4.8.1.
Current Supply	Standard - capable of providing levels stated in 4.8.1.

NOTE:
1/ Must be calibrated to the requirements of MIL-STD-45662 prior to use.

4.8 Methods of inspection. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the Government procuring agency for approval prior to use (see 4.7.2.2).

4.8.1 Performance tests. The following tests and examinations shall be performed to demonstrate compliance with

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3.4.2. The PWBA shall be connected to the circuit specified in Figure 7. Apply power by turning ON S1 and leaving on until all tests have been completed. All other switches shall be OFF unless otherwise specified in 4.8.1.1 through 4.8.1.3.

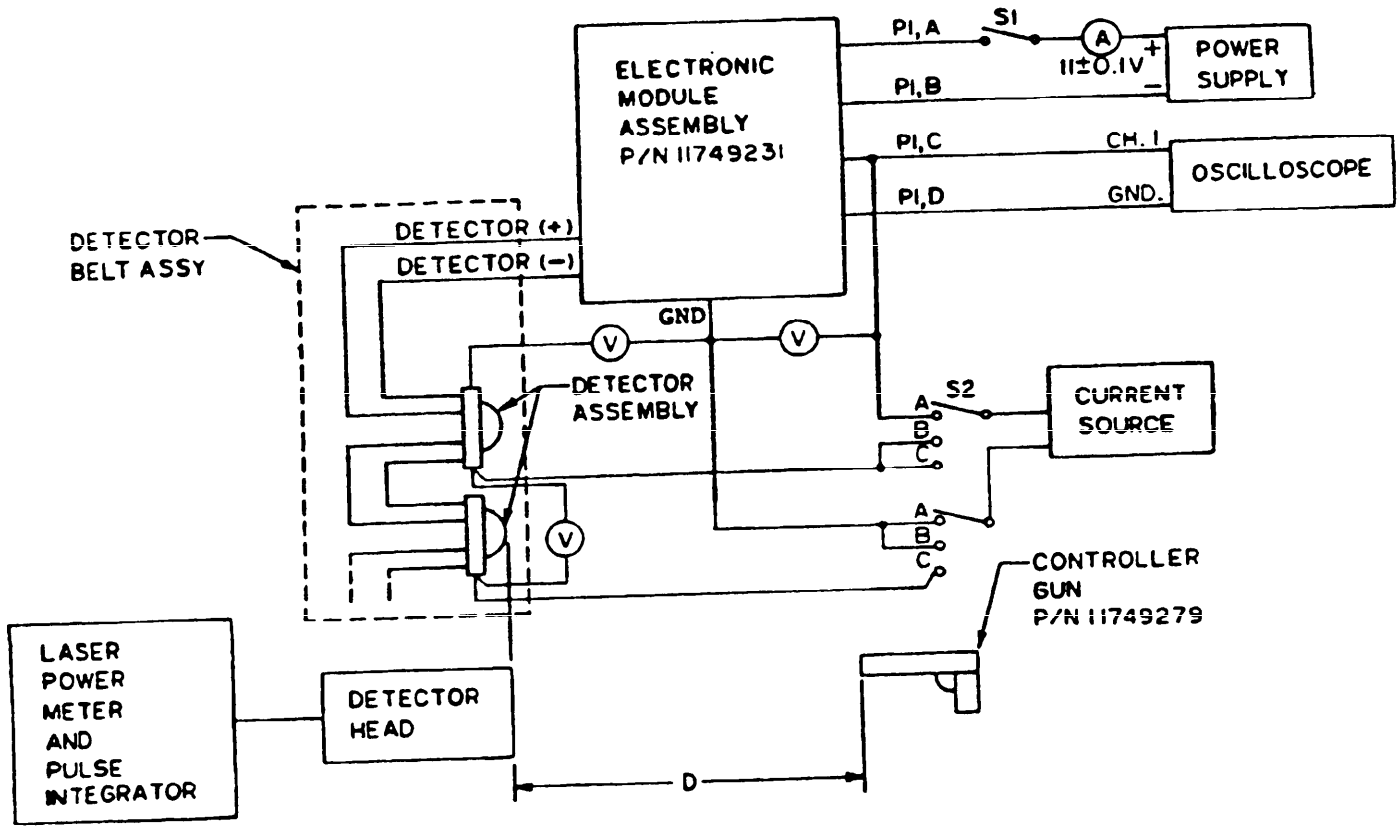


FIGURE 7. Test Circuit for Detector Belt Assembly.

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4.8.1.1 Current drain. Verify that the quiescent current meets the requirements of 3.4.2.1.

4.8.1.2 Detection Sensitivity. To verify the detection sensitivity requirement of 3.4.2.2, proceed as follows: Power the laser power meter and detector head (each calibrated to 904 nm) and the controller gun, and vary the distance D between the detector head and controller gun in order to meet the total collected energy requirements of Table II. Replace the detector head with the detector belt assembly at the same distance D and in full sunlight illumination or equivalent (12.4 lumens/cm^2), stimulate each detector by transmitting bits normal $\pm$ 10° to each detector. The detection sensitivity requirement will be met when the coded pulse rate seen by the oscilloscope is at least 50% the 3.1 kHz controller gun transmission rate. A typical coded pulse shall be as shown in Figure 2.

4.8.1.3 Grounding. To verify the grounding requirement of 3.4.2.3, proceed as follows: apply a minimum current of 0.5 A at S2 in position A and verify the resistance readings (voltage divided by two to give resistance) correspond with the grounding requirements using the Kelvin (4 wire) technique. Repeat the above procedure for S2 positions B and C and verify the corresponding grounding requirements.

4.9 Environmental tests.

a. Unless otherwise specified, environmental tests shall be conducted in accordance with the test conditions and general test performance guidance of MIL-STD-810. Test facilities, chambers and apparatus used in conducting the test required by this specification shall be capable of meeting the conditions specified.

b. Where necessary in the following tests, it shall be permissible to temporarily open chamber doors to permit access to and manipulation of test equipment and sample.

4.9.1 High temperature. This satisfies the storage and operating conditions of high temperature requirements as specified in 3.4.3.

4.9.1.1 Storage and operating. The assembly shall be tested in accordance with MIL-STD-810, Method 501.2, Section II, Procedures I and II, with the following modifications:

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a. Step 1 - Place the test item, configured for high temperature test, into the temperature chamber.

b. Step 2 - Raise the internal temperature to 62.8°C (+145°F).

c. Step 3 - Maintain the internal chamber temperature for a period of 4 hours while insuring the relative humidity is not in excess of 15 percent.

d. Step 4 - Operate the test item until the item is stabilized and perform the tests required in 4.8.1.

e. Step 5 - Return the test item, nonoperating, to standard ambient conditions and stabilize.

f. Step 6 - Visually inspect for physical damage and in the absence of physical damage, perform the tests required in 4.8.1.

4.9.2 Low temperature. This satisfies the storage and operating conditions of low temperature requirements as specified in 3.4.3.

4.9.2.1 Storage and operating. The assembly shall be tested in accordance with MIL-STD-810, Method 502.2 using the following procedure:

a. Step 1 - Place the test item, configured for low temperature test, into the temperature chamber.

b. Step 2 - Lower the internal chamber temperature to -34.4°C (-30°F) and maintain for a period of 24 hours after stabilization.

c. Step 3 - Adjust the internal chamber temperature to -31.7°C (-25°F) and maintain until temperature stabilization of the test item is reached.

d. Step 4 - Operate the test item until the item is stabilized and perform the tests required in 4.8.1.

e. Step 5 - Return the test item, nonoperating, to standard ambient conditions and stabilize.

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f. Step 6 - Visually inspect for physical damage and in the absence of physical damage, perform the tests required in 4.8.1.

4.9.3 Shock. The assembly, when mounted to simulate its next higher assembly, shall be tested in accordance with MIL-STD-810, Method 516.3, Section II, Procedure I, with the following modifications:

a. Step 3 - the assembly shall be in a non-operational mode.

b. The step 3 test transient shall be in accordance with 3.4.3.3.

c. Step 5 - the assembly inspection and functional verification shall not be conducted until shock has been completed in all three axes.

d. After shock testing the assembly shall not be damaged and shall be subjected to the performance tests specified in 4.8.1.

4.9.4 Vibration. The assembly, when mounted to simulate its next higher assembly, shall be tested in accordance with MIL-STD-810, Method 514.3, Section II, Procedure I, for a duration of 60 minutes per axis with the following modifications:

a. Step 4 - mount the test item on vibration equipment.

b. Step 5 - expose the assembly to test level and durations as specified in 3.4.3.4.

c. Step 6 - does not apply.

d. Step 7 - the assembly inspection and functional verification shall not be conducted until vibration has been completed in all three axes.

e. After vibration testing the assembly shall not be damaged and shall be subjected to the performance tests specified in 4.8.1.

4.9.5 Low pressure. The assembly shall be tested in accordance with MIL-STD-810, Method 502, Procedure I and meet the requirements of 3.4.3.5.

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4.9.6 Humidity. The assembly shall be tested in accordance with MIL-STD-810, Method 507, and meet the requirements of 3.4.3.6 using the following procedure:

a. Step 1 - Prior to starting the test, the internal chamber temperature shall be at standard ambient with uncontrolled humidity.

b. Step 2 - Gradually raise internal chamber temperature to 62.8°C (145°F) and the relative humidity to 95 +5 -3 percent over a period of 2 hours.

c. Step 3 - Maintain conditions of step 2 for not less than 6 hours.

d. Step 4 - Maintain 85 percent, or greater, relative humidity and reduce internal chamber temperature in 16 hours to 30°C (86°F).

e. Step 5 - Repeat steps 2, 3, and 4 for a total of 10 cycles (not less than 240 hours). Figure 5 is an outline of the humidity cycle for this procedure.

f. Step 6 - At the end of the tenth cycle, while still at 30°C (86°F) and 85 percent relative humidity, visually inspect for physical damage and in the absence of physical damage, perform the tests required in 4.8.1.

4.9.7 Rain. The assembly shall be tested in accordance with MIL-STD-810, Method 506, and meet the requirements of 3.4.3.7 as outlined in 4.9.7.1.

4.9.7.1 Rain test procedure. The test item shall be placed in the chamber in its normal operation position. The test item shall be exposed to a simulated rain at a rate of 2±0.5 inches per hour for 10 minutes with a water temperature between 52°F and 95°F. The rate of rainfall shall then be raised to 5 ±1 inches per hour and held at this rate for 5 minutes. The rate shall then be reduced to 2±0.5 inches per hour for the next 15 minutes. Starting 5 minutes after the initiation of the rain, the wind source shall be turned on and adjusted to produce a horizontal wind velocity of 40 miles per hour (3,500 feet per minute). The wind source shall be maintained at this velocity for 15 minutes after which the wind source shall be turned off. At the end of the 30 minute period, remove the assembly from the chamber and visually inspect for physical damage and in absence of physical damage, perform the tests required 4.8.1.

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4.9.8 Dust. The assembly shall be tested in accordance with MIL-STD-810, Method 510, and meet the requirements of 3.4.3.8 as outlined in 4.9.8.1.

4.9.8.1 Dust test procedure. Position the test item as near the center of the chamber as practicable. If more than one item is being tested, there shall be a minimum clearance of 4 inches between surfaces of test items or any other material or object capable of furnishing protection. Also, no surface of the test item shall be closer than 4 inches from any wall of the test chamber. Orient the item so as to expose the most critical or vulnerable parts to the dust stream.

a. Step 1 - Set the chamber controls to maintain an internal chamber temperature of 23°C (73°F) and a relative humidity of less than 22 percent. Adjust the air velocity to 1,750 ±250 feet per minute. Adjust the dust feeder to control the dust concentration at 0.3 ±0.2 grams per cubic foot. With the test item nonoperating, maintain these conditions for 6 hours.

b. Step 2 - Stop the dust feed and reduce the air velocity to 300 ±200 feet per minute. Raise the internal chamber air temperature to 62.8°C (145°F). Hold these conditions 16 hours.

c. Step 3 - While holding chamber temperature at 62.8°C (145°F) adjust the air velocity to 1,750 ±250 fpm. Adjust the dust feeder to control the dust concentration at 0.3 ±0.2 gms per cubic foot. Unless otherwise specified, with the test item nonoperating, maintain these conditions for 6 hours.

d. Step 4 - Turn off all chamber controls and allow the test item to return to standard ambient conditions. Remove accumulated dust from the test item by brushing, wiping, or shaking, care being taken to avoid introduction of additional dust into the test item. Dust shall not be removed by either air blast or vacuum cleaning.

e. Step 5 - Visually inspect for physical damage and in the absence of physical damage, perform the tests required in 4.8.1.

4.9.9 Salt fog. The assembly shall be tested in accordance with MIL-STD-810, Method 509, and meet the requirements of

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3.4.3.9 as outlined in 4.9.9.1.

4.9.9.1 Salt fog test procedure. The test item shall be placed in the test chamber and exposed to the salt fog for a period of 48 hours. At the end of the exposure period, unless otherwise specified, visually inspect for corrosion and in the absence of corrosion perform the tests required in 4.8.1. If necessary to aid in examination, a gentle wash in running water not warmer than 38°C (100°F) may be used. The test item shall then be stored in an ambient atmosphere for 48 hours. At the end of the drying period, visually inspect for physical damage and in the absence of physical damage, perform the tests required in 4.8.1.

4.9.10 Electromagnetic Interference (EMI). The belt assembly shall be tested in accordance with MIL-STD-462, and shall meet the requirements specified in 3.4.3.10. The belt assembly shall be tested using the following set-up.

4.9.10.1 Electromagnetic Interference equipment set-up. The belt assembly shall be operated by the console, simulator system. The test sample shall be provided optical stimulation by a MILES transmitter, such as the controller gun. The test sample will be mounted on a ground plane in such a way as to simulate its installation upon the hull of a vehicle. The console, simulator system shall be connected to the screen room ground plane by a low impedance connection. The test sample shall be connected to the console, simulator system.

4.10 Certifications. Certification shall be provided that all units have been produced in accordance with the requirements of 3.5 and packaged in accordance with the requirements of 5.1.

4.11 Reliability demonstration. A reliability demonstration shall be performed in accordance with the requirements of MIL-STD-781, fixed ground test conditions, Thermal stress b, and meet the requirements specified in 3.6.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with corresponding special packaging instruction: SPI-P11749230, Segment No. 1 - CVLD; SPI-P11749238, Segment No. 2 - CVLD; SPI-P11749300, Segment No. 3 - CVLD; SPI-P 11749296, Segment No. 4 - CVLD; SPI-P11835890, Segment No. 5 - CVLD; and SPI-P11836200, Segment No. 6 - CVLD. The level of

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protection shall be A, B, or C, as defined by the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Detector Belt Assemblies are part of the Multiple Integrated Laser Engagement System. The assemblies included are: Segment No. 1 - CVLD, P/N 11749230; Segment No. 2 - CVLD, P/N 11749238; Segment No. 3 - CVLD, P/N 11749300; Segment No. 4 - CVLD, P/N 11749296; Segment No. 5 - CVLD, P/N 11835890; Segment No. 6 - CVLD, P/N 11836200.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number and date of this specification (see page 1).

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Special tests and examinations required (see 4.2); the type of inspection required; and the activity responsible for testing, and its facility (see 4.1).

d. Whether a first article sample is required (see 3.2), and if so, pertinent details (see 4.4) including disposition of the first article sample.

e. Certification of conformance for each lot or shipment of product.

f. Whether Government-Loaned Property (Government Furnished inspection/test equipment) is available or will the contractor be required to furnish inspection equipment (see 4.7).

g. Applicable level(s) of preservation, packaging, packing and marking instructions (see 5.1).

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6.3 Submission of contractor inspection equipment designs for approval. Unless otherwise specified, submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423, in the contract. The letter of submittal shall specify, the contractor; contract number; title, number and revision of this specification; item nomenclature; classification of defect and test paragraph.

6.4 Definitions.

6.4.1 MILES. Multiple Integrated Laser Engagement System.

6.4.2 SME. Standard Measuring Equipment.

6.4.3 GaAs. Gallium Arsenide.

6.5 Drawings. Drawings listed in Section 2 of this specification under heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.6 Noncompliance. If a sample fails to pass group B inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Group A inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group B inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

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6.7 Government - loaned property. Unless otherwise specified (see 6.2), the contracting officer should arrange to loan the following Government furnished Inspection/Test equipment specified below:

P/N	Title	Ref. Paragraph
11749279	Controller Gun	4.8.1.2 and 4.9.10
11749549	Console, Simulator System	4.9.10

6.8 Subject term (key word) listing.

Controller Gun
Fire Control
Laser
Training Devices

6.9 Supersession. The predecessor of MIL-D-70860 (AR) was Prime Item Specification 11749550, it was used for the US Army AMCCOM and was not forwarded to the Naval Publications and Forms Center for publication.

Custodians:
Army - AR

Preparing activity:
Army - AR

(Project 1265-A024)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revisio letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-D-70860 (AR)	14 FEBRUARY 1992

3. DOCUMENT TITLE: DETECTOR BELT ASSEMBLY, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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